

# Mirza Uzaif Baig

*Developer, Builder, Problem Solver*

Kalaburagi, Karnataka, India

B.E. in Artificial Intelligence & Machine Learning

Poojya Doddappa Appa College of Engineering (2023–2027)

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Mirza-Uzaif-Baig

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Portfolio

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## PROFESSIONAL SUMMARY

AI & ML undergraduate and software engineer focused on building scalable AI-powered and distributed applications. Hands-on experience across full-stack development, backend engineering, system design, cloud infrastructure, computer vision, and mobile applications. Co-Founder & Tech Lead at Tuguduk, responsible for technical architecture, application design, backend logic, infrastructure, deployment, and end-to-end engineering. Experienced with Python, JavaScript, Node.js, React Native, MySQL, AWS, Docker, Linux, and modern backend infrastructure.

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## EXPERIENCE

### Tuguduk: Co-Founder & Tech Lead

May 2026 – Present | NAIN 2.0, Government of Karnataka

- Lead the complete technical development of Tuguduk, a multi-service logistics, mobility, fuel intelligence, and delivery platform.
- Own the technical architecture, system design, application structure, backend logic, API design, database architecture, infrastructure, and deployment strategy.
- Design and implement scalable service workflows for transportation, logistics, fuel availability intelligence, and on-demand delivery.
- Develop the mobile application using React Native and build backend services to support centralized business logic and service orchestration.
- Design infrastructure and deployment architecture using VPS, Linux, Nginx, Docker, routing, and reverse-proxy configurations.
- Work on scalability and reliability concepts including load balancing, caching with Redis, database optimization, data replication, and horizontal scaling.
- Design service boundaries, request routing, data flows, and communication between application components.
- Lead technical problem-solving across the complete development lifecycle, from architecture and database design to implementation, deployment, and maintenance.
- Develop the platform under NAIN 2.0 with support from the Government of Karnataka.

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## TECHNICAL SKILLS

**Programming Languages:** Python, C++, JavaScript

**AI & Automation:** Artificial Intelligence, Agentic AI, RAG Workflows, n8n

**System Design & Architecture:** System Design, Distributed Systems, Scalability, Load Balancing, Routing, Caching, Database Sharding, Data Replication, Service Architecture

**Backend & APIs:** Node.js, Express.js, REST APIs, API Integration, Backend Processing

**Cloud & Infrastructure:** AWS (EC2, S3, IAM, VPC, Lambda), VPS, Docker, Linux, Nginx, Cloud Deployment

**Databases & Data Systems:** MySQL, SQLite, Redis, Database Design, Query Optimization, Data Replication, Sharding

**Mobile & Web Development:** React Native, HTML5, CSS3, JavaScript, Responsive Web Design

**DevOps & Tools:** Docker, Git, GitHub, Bash, Linux, VS Code, Jupyter Notebook, Google Colab

**Core Computer Science:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Distributed Systems

## EDUCATION

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### B.E. in Artificial Intelligence & Machine Learning

CGPA: 7.56 (6th Semester)

Poojya Doddappa Appa College of Engineering, Kalaburagi

2023 – 2027

### Pre-University (12th Grade)

Score: 92.5%

Al-Sharay PU College, Kalaburagi

March 2023

## PROJECTS

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### RuleCop - AI-Based University Violation Detection System

2025

*Python, OpenCV, YOLOv8, MySQL, Web Dashboard*

- Developed a real-time computer vision system capable of processing 25 FPS video streams with more than 88% detection accuracy.
- Designed a structured MySQL database schema to store violation logs and analytics.
- Built a web-based administrative dashboard to monitor detection results and system output.
- Implemented backend processing pipelines that handle video ingestion, detection, and data storage.

Research Paper

GitHub

### BrainShh - AI Mental Wellness Web Application

2026

*HTML5, CSS3, JavaScript, Node.js, AWS Elastic Beanstalk*

- Built an AI-powered mental wellness web app with mood tracking, guided breathing, and journaling features.
- Developed coping strategy modules and supportive conversational flows focused on managing day-to-day stress.
- Designed backend service logic for scalable session handling and cloud-ready deployment.

Screenshots

### PayOff - Offline Token-Based Payment Application

2026 – Present

*React Native, Expo, Node.js, Express.js, Socket.IO, Python, MySQL – In Progress*

- Developing an offline-capable payment application using token-based transactions to support payments without continuous internet connectivity.
- Building a React Native and Expo mobile client with Node.js, Express.js, and Socket.IO backend services for transaction processing and communication.
- Design a local transaction ledger and synchronization workflow to preserve transaction records and reconcile data when connectivity is restored.
- Implemented token generation, validation, transaction processing, and secure synchronization between application components.

GitHub

## CERTIFICATIONS

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Cloud Computing - NPTEL

May 2026

Data Analytics - Head Held High

Apr 2026

HTML & CSS - Certiport

Jul 2024